

Zain Khan

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Computer Engineering

– Awarded over **\$40,000** in merit-based scholarships, including the **University of Waterloo Presidential Scholarship**.

Waterloo, ON

Sep. 2025 – Apr. 2030

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Ruby, C/C++, Java, SQL, HTML/CSS

Frameworks: React, Redux, Next.js, Node.js, Express, Django, FastAPI, Three.js, TensorFlow, OpenCV, Tailwind CSS

Infrastructure: PostgreSQL, MySQL, MongoDB, Supabase, AWS, Docker, Linux, Git, Vercel

Protocols: HTTP/REST, GraphQL, gRPC, WebRTC, WebSockets, OAuth 2.0

EXPERIENCE

Leap Tools

Software Development Intern, Bespoke Team

Toronto, ON

May 2026 – Aug. 2026

- Shipped shopper-facing commerce experiences on the Roomvo AR visualizer in TypeScript, React/Redux, and Next.js, enabling buyers across **80+ countries** to preview flooring, rugs, and tiles in real room photos before purchase.
- Prototyped a Surface Designer that renders custom (bespoke) surface patterns as vectorized frontend stencils, demonstrating a client-side alternative to the existing C++ algorithmic pattern-generation backend that eliminates per-render server round-trips and enables instant, interactive pattern editing directly in the browser.
- Traced production requests through Python/Django microservices on PostgreSQL and AWS, implementing HTTP and gRPC APIs powering commerce for **Fortune 500** retailers including **Home Depot**.

WATonomous

Robotics Software Engineer, EVE Autonomy

Waterloo, ON

Jan. 2026 – Present

- Developed concurrent autonomous robotics software in C++ on ROS 2 Humble, coordinating **5+ real-time nodes** across perception, planning, and control subsystems for an autonomous delivery platform.
- Implemented costmaps, A* path planning, and Pure Pursuit control, achieving stable point-to-point navigation in cluttered indoor environments with dynamic obstacle avoidance.

Kasperia

Software Developer (Contract), Frontend & Data

Remote

Jun. 2023 – Sep. 2025

- Architected reusable React components with modular CSS across the client-facing web application, improving page load performance by **30%** and reducing maintenance overhead by **25%**.
- Built automated data pipelines using JavaScript, SQL, and REST APIs to consolidate reporting workflows, cutting manual reporting time by **40%** and powering real-time KPI dashboards.

PROJECTS

otto  | *Next.js, TypeScript, Python, LiveKit WebRTC, Deepgram, Gemini 2.5 Flash, Supabase*

- Shipped an AI voice agent that turns natural speech into automated actions across GitHub, Gmail, and Google Calendar; awarded **1st Place — LiveKit Track** at NexHacks.
- Engineered a LiveKit WebRTC pipeline with Deepgram VAD turn detection and Gemini 2.5 Flash Realtime native audio inference, achieving **sub-second** end-to-end voice-to-action latency.
- Integrated OAuth 2.0 auto-refresh via Supabase Auth and compressed agent context with TTC Bear-1, reducing per-request token usage from **4000 → 1800 (55%)**.

toothfairy  | *Next.js, TypeScript, FastAPI, Python, TensorFlow, OpenCV, Gemini 2.5 Flash, Moorcheh AI*

- Built a production-grade dental X-ray segmentation pipeline using a U-Net (TensorFlow/Keras) with OpenCV connected component analysis and morphological post-processing, isolating **28+ individual teeth** from panoramic radiographs.
- Integrated Moorcheh AI for per-patient RAG with isolated memory namespaces and semantic search across visits, enabling longitudinal retrieval for clinical history and diagnostic correlation.
- Orchestrated **3 AI copilots** using Gemini 2.5 Flash for structured JSON extraction and Gemini Vision for pathology detection with confidence scoring across **10 dental conditions**.

extgen | *React, TypeScript, Chrome Extensions, FastAPI, Python, Agentverse, uAgents, Gemini, WebSockets*

- Built a Chrome side panel that turns natural-language requests into installable Manifest V3 extensions, generating content scripts, CSS, validated project files, ZIP packages, and load-unpacked instructions.
- Implemented Edit DOM mode with Cmd/Ctrl-hover element selection, ordered edit chips, live DOM operations, and one-click export from temporary page edits into a permanent extension.
- Designed an Agentverse/uAgents orchestration pipeline with Architect, RAG, Codegen, Validator, and Packager roles, combining deterministic intent templates with LLM generation and per-site DOM bootstrap retrieval.